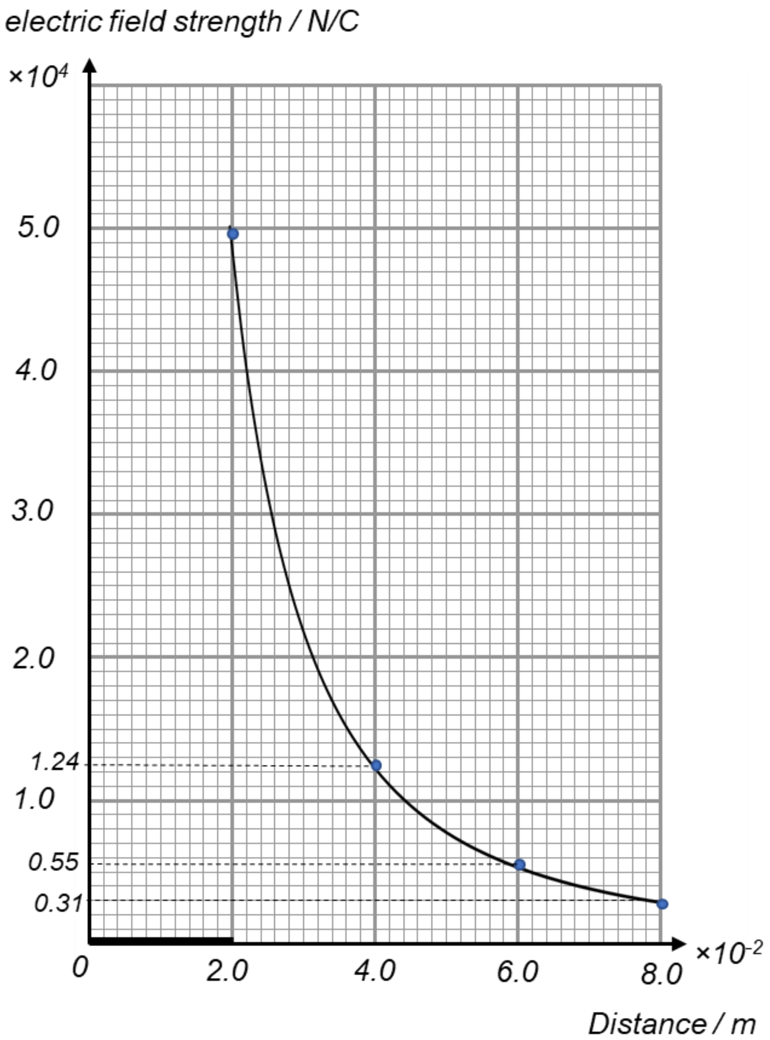
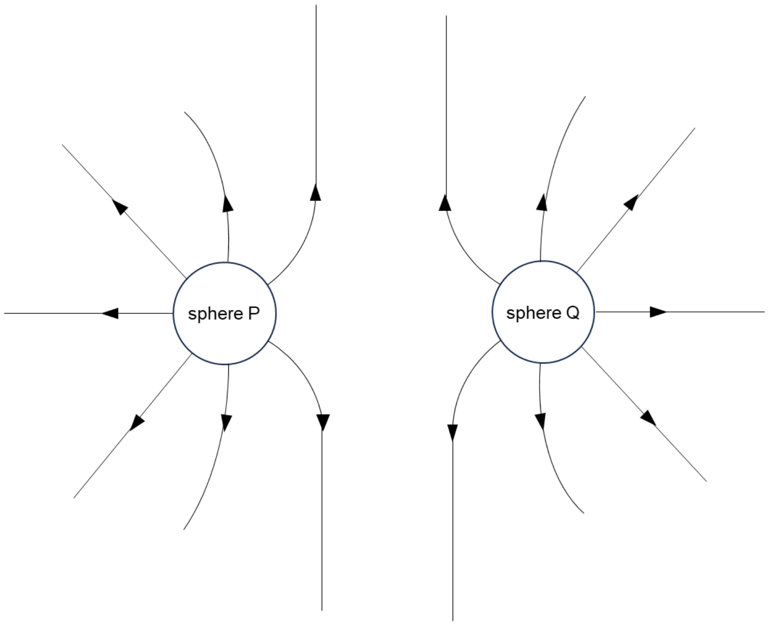
2	
(a)	Electric field strength E at a point is defined as the electric force exerted per unit positive charge
	on a small stationary test charge placed at that point in the electric field.
(b)(i)	$E = \dfrac{Q}{4\pi \varepsilon_0 r^2}$ $= \dfrac{2.2\times10^{-9}}{4\pi(8.85\times10^{-12})(\frac{4.0\times10^{-2}}{2})^2}$
	= 4.94 $\times10^{4}$ N C^{-1}
(ii)	

(c)(i)	
(ii)	Electric field strength is the negative potential gradient
	Gradient points in the direction of the greatest change of potential which is the (opposite) direction of the electric field. There is no component of the electric field strength (electric force) parallel to the equipotential line as this will cause a change in electric potential